

**2024-2025 Fall**  
**MAT123 Makeup**  
**31/01/2025**

1.
  - a. Find  $\int \frac{\sin^3 x}{\sqrt{\cos x}} dx$ .
  - b. Find  $\int \frac{dx}{x^3 - 4x^2 + 3x}$ .
  - c. Evaluate  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\pi/2}^x \ln(\sin t) dt}{\sin x - 1}$ .
  - d. Evaluate the improper integral  $\int_0^2 \frac{dx}{(x-1)^{2/3}}$ .
2. Consider the region  $R$  bounded by the curves  $y = \arctan x$ ,  $y = \ln x$  and the lines  $x = \frac{1}{\sqrt{3}}$  and  $x = 1$ .
  - a. Sketch the region and find the area of the  $R$ .
  - b. Write a definite integral (do not evaluate) by using the Washer Method, which gives the volume of a solid obtained by rotating the region  $R$  about the  $y$ -axis.
  - c. Write a definite integral (do not evaluate) by using the Cylindrical Shell Method, which gives the volume of a solid obtained by rotating the region  $R$  about the line  $y = 2$ .
3. Determine whether each series is convergent or divergent. Explain your answer.
  - a.  $\sum_{n=0}^{\infty} \frac{(-3)^{n+1}}{2^{3n}}$
  - b.  $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$
4. Find the Taylor series of the function  $f(x) = \ln x$  at  $c = 1$  and determine the interval of convergence.

1. a. Use the method of partial fraction decomposition.

$$\int \frac{\sin^3 x}{\sqrt{\cos x}} dx = \int \frac{(1 - \cos^2 x) \cdot \sin x}{\sqrt{\cos x}} dx$$

Let  $u = \cos x$ , then  $du = -\sin x dx$ .

$$\begin{aligned} \int \frac{(1 - \cos^2 x) \sin x}{\sqrt{\cos x}} dx &= \int -\frac{(1 - u^2)}{\sqrt{u}} du = \int \left( u^{3/2} - \frac{1}{\sqrt{u}} \right) du \\ &= \frac{2}{5} u^{5/2} - 2\sqrt{u} + c \\ &= \boxed{\frac{2}{5} (\cos x)^{5/2} - 2\sqrt{\cos x} + C, \quad C \in \mathbb{R}} \end{aligned}$$

- b. Use the method of partial fraction decomposition.

$$\int \frac{dx}{x^3 - 4x^2 + 3x} = \int \frac{dx}{x(x-3)(x-1)} = \int \left( \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x-1} \right) dx$$

$$\begin{aligned} A(x-3)(x-1) + Bx(x-1) + Cx(x-3) &= 1 \\ x^2(A+B+C) + x(-4A-B-3C) + 3A &= 1 \end{aligned}$$

Equate the coefficients of like terms.

$$\left. \begin{aligned} x^2(A+B+C) &= 0 \\ x(-4A-B-3C) &= 0 \\ 3A &= 1 \end{aligned} \right\} \rightarrow A = \frac{1}{3}, \quad \left. \begin{aligned} B+C &= -\frac{1}{3} \\ B+3C &= -\frac{4}{3} \end{aligned} \right\} \rightarrow B = \frac{1}{6}, \quad C = -\frac{1}{2}$$

Rewrite the integral.

$$\begin{aligned} \int \left( \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x-1} \right) dx &= \int \left( \frac{1}{3x} + \frac{1}{6(x-3)} - \frac{1}{2(x-1)} \right) dx \\ &= \boxed{\frac{1}{3} \ln |x| + \frac{1}{6} \ln |x-3| - \frac{1}{2} \ln |x-1| + C_0, \quad C_0 \in \mathbb{R}} \end{aligned}$$

- c. The limit is in the indeterminate form  $0/0$ . Apply L'Hôpital's rule to eliminate the indeterminate form.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_{\pi/2}^x \ln(\sin t) dt}{\sin x - 1} \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{d}{dx} \int_{\pi/2}^x \ln(\sin t) dt}{\cos x}$$

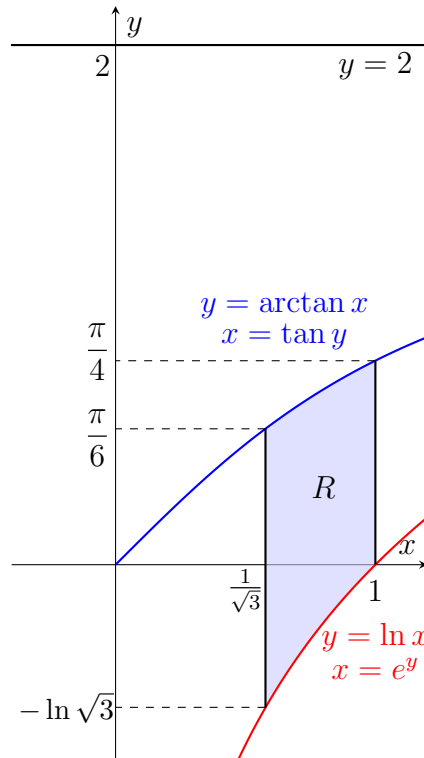
By the Fundamental Theorem of Calculus, we may rewrite the limit as follows.

$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{d}{dx} \int_{\pi/2}^x \ln(\sin t) dt}{\cos x} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{\cos x} \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\sin x} \cdot \cos x}{-\sin x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{-\sin^2 x} \\ &= -\frac{\cos \frac{\pi}{2}}{\sin^2 \frac{\pi}{2}} = \boxed{0} \end{aligned}$$

d. Take the limit since this is an improper integral.

$$\begin{aligned} \int_0^2 \frac{dx}{(x-1)^{2/3}} &= \lim_{R \rightarrow 1^-} \int_0^R \frac{dx}{(x-1)^{2/3}} + \lim_{P \rightarrow 1^+} \int_P^2 \frac{dx}{(x-1)^{2/3}} \\ &= \lim_{R \rightarrow 1^-} 3(x-1)^{1/3} \Big|_0^R + \lim_{P \rightarrow 1^+} 3(x-1)^{1/3} \Big|_P^2 \\ &= 3 \lim_{R \rightarrow 1^-} ((R-1)^{1/3} - (-1)) + 3 \lim_{P \rightarrow 1^+} (1 - (P-1)^{1/3}) = \boxed{6} \end{aligned}$$

2. a.



$$A = \int_{1/\sqrt{3}}^1 (\arctan x - \ln x) dx = \int_{1/\sqrt{3}}^1 \arctan x dx - \int_{1/\sqrt{3}}^1 \ln x dx \quad (1)$$

Calculate the first integral in (??) by integration by parts.

$$\left. \begin{array}{l} u = \arctan x \implies du = \frac{1}{x^2 + 1} dx \\ dv = dx \implies v = x \end{array} \right\} \rightarrow \int u dv = uv - \int v du$$

$$\begin{aligned} \int_{1/\sqrt{3}}^1 \arctan x dx &= x \arctan x \Big|_{1/\sqrt{3}}^1 - \int_{1/\sqrt{3}}^1 \frac{x}{x^2 + 1} dx \\ &= \left( x \arctan x - \frac{1}{2} \ln |x^2 + 1| \right) \Big|_{1/\sqrt{3}}^1 \\ &= \left( \frac{\pi}{4} - \frac{\ln 2}{2} \right) - \left( \frac{\pi\sqrt{3}}{18} - \frac{1}{2} \cdot \ln \left( \frac{4}{3} \right) \right) \\ &= \frac{\pi(9 - 2\sqrt{3})}{36} + \frac{1}{2} \cdot \ln \frac{2}{3} \end{aligned}$$

Calculate the second integral in (??) by integration by parts.

$$\left. \begin{array}{l} u = \ln x \implies du = \frac{1}{x} dx \\ dv = dx \implies v = x \end{array} \right\} \rightarrow \int u dv = uv - \int v du$$

$$\begin{aligned} \int_{1/\sqrt{3}}^1 \ln x dx &= x \ln x \Big|_{1/\sqrt{3}}^1 - \int_{1/\sqrt{3}}^1 dx = (x \ln x - x) \Big|_{1/\sqrt{3}}^1 \\ &= (0 - 1) - \left( -\frac{\ln \sqrt{3}}{\sqrt{3}} - \frac{1}{\sqrt{3}} \right) = \frac{\sqrt{3} \ln \sqrt{3} + \sqrt{3} - 3}{3} \end{aligned}$$

We have

$$A = \frac{\pi(9 - 2\sqrt{3})}{36} + \frac{1}{2} \cdot \ln \frac{2}{3} - \frac{\sqrt{3} \ln \sqrt{3} + \sqrt{3} - 3}{3}.$$

b.

$$\begin{aligned} &\int_{-\ln \sqrt{3}}^0 \pi \left[ (e^y)^2 - \left( \frac{1}{\sqrt{3}} \right)^2 \right] dy + \int_0^{\pi/6} \pi \left[ (1)^2 - \left( \frac{1}{\sqrt{3}} \right)^2 \right] dy \\ &+ \int_{\pi/6}^{\pi/4} \pi \left[ (1)^2 - (\tan y)^2 \right] dy \end{aligned}$$

c.

$$\int_{-\ln \sqrt{3}}^0 2\pi(2-y) \left( e^y - \frac{1}{\sqrt{3}} \right) dy + \int_0^{\pi/6} 2\pi(2-y) \left( 1 - \frac{1}{\sqrt{3}} \right) dy + \int_{\pi/6}^{\pi/4} 2\pi(2-y)(1 - \tan y) dy$$

3. a. Rewrite the sum.

$$\sum_{n=0}^{\infty} \frac{(-3)^{n+1}}{2^{3n}} = \sum_{n=0}^{\infty} \frac{-3 \cdot (-3)^n}{8^n} = -3 \sum_{n=0}^{\infty} \left( -\frac{3}{8} \right)^n$$

This is a geometric series where  $r = -\frac{3}{8}$ .  $|r| = \frac{3}{8} < 1$ . Therefore, the series

$$\sum_{n=0}^{\infty} \frac{(-3)^{n+1}}{2^{3n}} \text{ converges.}$$

b. Take  $f(x) = \frac{1}{x(\ln x)^2}$ .  $f$  is positive and decreasing for  $x \geq 2$  because  $x$  and  $(\ln x)^2$  are positive and increasing for  $x \geq 2$ .  $x$  is a polynomial which is defined everywhere and  $(\ln x)^2$  is continuous for  $x \geq 2$ . Since we took into account every criterion, we may apply the Integral Test. Handle the improper integrals by taking the limit.

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \lim_{R \rightarrow \infty} \int_2^R \frac{dx}{x(\ln x)^2} = \lim_{R \rightarrow \infty} \left[ -\frac{1}{\ln x} \right]_2^R = \lim_{R \rightarrow \infty} \left[ -\frac{1}{\ln R} + \frac{1}{\ln 2} \right] = \frac{1}{\ln 2}$$

Since the integral converges, the series  $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$  also converges.

4. The Taylor series of  $f$  at  $c = 1$  is as follows.

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(1)}{k!} (x-1)^k$$

Find  $f(1)$ ,  $f'(1)$ ,  $f''(1)$ ,  $f'''(1)$ ,  $f^{(4)}(1)$  to look for the pattern.

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2}, \quad f'''(x) = \frac{2}{x^3}, \quad f^{(4)}(x) = -\frac{6}{x^4}$$

$$f(1) = 0, \quad f'(1) = 1, \quad f''(1) = -1, \quad f'''(1) = 2, \quad f^{(4)}(1) = -6$$

This is an alternating sequence where the coefficient of each term is the factorial of the subsequent number starting from 0 except for  $k = 0$ , that is, the first term of the series. At  $k = 0$ , the first term is 0. So,

$$f^k(1) = \begin{cases} (-1)^{k-1} \cdot (k-1)!, & \text{if } k > 0 \\ 0, & \text{if } k = 0 \end{cases}.$$

$$\begin{aligned}\sum_{k=0}^{\infty} \frac{f^{(k)}(1)}{k!} x^k &= 0 + \sum_{k=1}^{\infty} \frac{f^{(k)}(1)}{k!} (x-1)^k = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \cdot (k-1)!}{k \cdot (k-1)!} (x-1)^k \\ &= \boxed{\sum_{k=1}^{\infty} \frac{(-1)^{k-1} (x-1)^k}{k} = (x-1) - \frac{(x-1)^2}{2} + \dots}\end{aligned}$$

Now, determine the interval of convergence. Apply the Ratio Test.

$$\begin{aligned}\lim_{k \rightarrow \infty} \left| \frac{(-1)^k (x-1)^{k+1}}{k+1} \cdot \frac{k}{(-1)^{k-1} (x-1)^k} \right| &= \lim_{k \rightarrow \infty} \left| \frac{(x-1) \cdot k}{(k+1) \cdot (-1)} \right| \\ &= |x-1| \lim_{k \rightarrow \infty} \left| \frac{k}{k+1} \right| = |x-1|\end{aligned}$$

$$|x-1| < 1 \implies -1 < x-1 < 1 \implies 0 < x < 2 \quad (\text{convergent})$$

Investigate the convergence at the endpoints.

$$x = 0 \rightarrow \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot (-1)^k}{k} = \sum_{k=1}^{\infty} \frac{(-1)^{2k} \cdot (-1)}{k} = - \sum_{k=1}^{\infty} \frac{1}{k}$$

This is a  $p$ -series with  $p = 1$ , for which the series diverges by the  $p$ -series Test. Try  $x = 2$ .

$$x = 2 \rightarrow \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

This is an alternating series. The non-alternating part, which is  $\frac{1}{k}$ , is nonincreasing for  $k \geq 1$  and it is positive. The limit at infinity is 0. By Leibniz's Alternating Series Test, the series converges.

The convergence set for the power series is  $\boxed{(0, 2]}$ .